

22 Anomalies

22.2 Transformation of the Measure: The Abelian Anomaly

We now turn to a calculation of anomalies of the sort relevant to π^0 decay. For this purpose we adopt Fujikawa's interpretation [6] of the anomaly as a symptom of the impossibility of defining a suitably invariant measure for integrations over fermionic field variables. Fujikawa's analysis of this problem was based on the use of path integrals in Euclidean space, and on an expansion of the fermionic variables of integration in eigenfunctions of the gauge-invariant Dirac operator, which is Hermitian in four-dimensional Euclidean space. Here we shall first present a less rigorous derivation, based on the familiar path integrals in Minkowski space, which will allow us to obtain the correct answer with a minimum of work. The Euclidean approach will be taken up briefly at the end of this section, and used to derive a famous index theorem.

We will start by evaluating the anomaly in the transformation of the measure under an arbitrary local matrix transformation $\psi(x) \rightarrow U(x)\psi(x)$ of a column $\psi_n(x)$ of massless complex spin 1/2 fermion fields that interact non-chirally with a set of gauge fields $A_\alpha^\mu(x)$ (such as the u and d quark fields interacting with the electromagnetic vector potential $A^\mu(x)$ in the problem of calculating the rate for $\pi^0 \rightarrow 2\gamma$). These are fermionic variables, so the measure is transformed not with the determinant of the transformation matrix, but with its inverse:

$$[d\psi][d\bar{\psi}] \longrightarrow (\text{Det } \mathcal{U} \text{ Det } \bar{\mathcal{U}})^{-1} [d\psi][d\bar{\psi}], \quad (22.2.1)$$

where

$$\mathcal{U}_{xn,ym} \equiv U(x)_{nm} \delta^4(x-y), \quad (22.2.2)$$

$$\bar{\mathcal{U}}_{xn,ym} \equiv [\gamma_4 U(x)^\dagger \gamma_4]_{nm} \delta^4(x-y), \quad (22.2.3)$$

and $\gamma_4 \equiv i\gamma^0$ is the matrix used to define $\bar{\psi} = \psi^\dagger \gamma_4$. The indices n, m run over flavor labels and Dirac spin indices.

The reader may wonder at this point why we bother to include factors of γ_4 in Eq. (22.2.3), since they just amount to a unitary transformation that should not affect the determinants. The answer is that in order for calculations to be meaningful we will find it necessary to regulate the sum over fermion modes in calculating propagators and determinants, and we shall find that the γ_4 factors do affect the regulated determinants. Whether or not we include the γ_4 factors thus depends on the method of regularization that is used to make these hand-waving manipulations meaningful. We include the factors of γ_4 because we wish to regulate in such a way as to preserve Lorentz invariance, and in the cases of interest here it is $\gamma_4 U(x)^\dagger \gamma_4$ rather than $U(x)^\dagger$ that transforms as a scalar.

First, let us consider the case where $U(x)$ is a unitary non-chiral transformation

$$U(x) = \exp[i\alpha(x)t], \quad (22.2.4)$$

with t an ordinary Hermitian matrix (not involving γ_5 , but not necessarily traceless) and $\alpha(x)$ an arbitrary real function of x . In this case $\mathcal{U}$ is pseudounitary:

$$\bar{\mathcal{U}}\mathcal{U} = \mathbf{1}, \quad (22.2.5)$$

so the measure is invariant under this sort of transformation. In particular, the symmetry under the gauge group itself, where t is one of the non-chiral generators t_α , is not spoiled by any anomalies.

Next, consider a local chiral transformation, with

$$U(x) = \exp[i\gamma_5\alpha(x)t], \quad (22.2.6)$$

with t again an ordinary Hermitian matrix and $\alpha(x)$ again an arbitrary real function of x . In this case $\mathcal{U}$ is pseudo-Hermitian:

$$\bar{\mathcal{U}} = \mathcal{U}. \quad (22.2.7)$$

The measure is not invariant under the chiral transformation; rather, we have

$$[d\psi][d\bar{\psi}] \longrightarrow (\text{Det } \mathcal{U})^{-2} [d\psi][d\bar{\psi}]. \quad (22.2.8)$$

Now let us specialize to the case of an *infinitesimal* local chiral transformation. Taking $\alpha(x)$ to be infinitesimal in Eq. (22.2.6), we have here

$$[\mathcal{U} - \mathbf{1}]_{nx,my} = i\alpha(x)[\gamma_5 t]_{nm}\delta^4(x-y). \quad (22.2.9)$$

Using the identity $\text{Det } M = \exp \text{Tr} \ln M$ and the limiting formula $\ln(1+x) \rightarrow x$ for $x \rightarrow 0$, the measure now has the transformation property

$$[d\psi][d\bar{\psi}] \longrightarrow \exp \left\{ i \int d^4x \alpha(x) \mathcal{A}(x) \right\} [d\psi][d\bar{\psi}], \quad (22.2.10)$$

where $\mathcal{A}$ is the anomaly function:

$$\mathcal{A}(x) \equiv -2 \text{Tr} \{ \gamma_5 t \} \delta^4(x-x), \quad (22.2.11)$$

with ‘Tr’ here denoting a trace over both Dirac and species indices. The measure $[d\psi][d\bar{\psi}]$ appears in the path integral weighted with a factor $\exp\{i \int d^4x \mathcal{L}(x)\}$, so the factor $\exp\{i \int d^4x \alpha(x) \mathcal{A}(x)\}$ in the transformation rule (22.2.10) for the measure has the same effect as if the Lagrangian density $\mathcal{L}(x)$ were not invariant under these transformations, but instead $\mathcal{L}(x) \rightarrow \mathcal{L}(x) + \alpha(x) \mathcal{A}(x)$. Hence when we use an effective Lagrangian with the fermions integrated out, to take account of the anomaly we must be sure to include a non-invariant term so that

$$\mathcal{L}_{\text{eff}}(x) \longrightarrow \mathcal{L}_{\text{eff}}(x) + \alpha(x) \mathcal{A}(x). \quad (22.2.12)$$

It now remains to calculate the anomaly function $\mathcal{A}(x)$.

At first sight it does not look like we can get any definite result for the anomaly. The delta-function is infinite, but the trace vanishes. To do better, we must introduce a regulator to make $\delta^4(x-x)$ meaningful. We can do this in a gauge-invariant manner by inserting a differential operator $f(-\not{D}_x^2/M^2)$ acting on the delta-function before taking its argument to zero:

$$\mathcal{A}(x) = -2 \left[\text{Tr} \left\{ \gamma_5 t f(-\not{D}_x^2/M^2) \right\} \delta^4(x-y) \right]_{y \rightarrow x}. \quad (22.2.13)$$

Here D_x is the Dirac differential operator in the presence of the gauge field $A_\alpha^\mu(x)$:

$$(D_x)_\mu \equiv \frac{\partial}{\partial x^\mu} - it_\alpha A_{\alpha\mu}(x). \quad (22.2.14)$$

Also, M is some large mass, eventually to be taken to infinity, and $f(s)$ is a smooth function, subject only to the condition¹ that as s goes from 0 to ∞ , $f(s)$ must drop smoothly from 1 to 0:

$$f(0) = 1, \quad f(\infty) = 0, \quad (22.2.15)$$

$$sf'(s) = 0 \quad \text{at} \quad s = 0 \text{ and } s = \infty. \quad (22.2.16)$$

Note that we do not take the regulator function to be a function of $\not{D}$, because we want to preserve gauge invariance, and we do not take it to be a function of $D^\mu D_\mu$, because we need it to regulate not only the determinant but also the fermion propagator $\not{D}^{-1}$.

To evaluate the expression (22.2.13), we use the Fourier representation of the delta-function, and write the anomaly function as

$$\begin{aligned} \mathcal{A}(x) &= -2 \int \frac{d^4 k}{(2\pi)^4} \left[\text{Tr} \left\{ \gamma_5 t f(-\not{D}_x^2/M^2) \right\} e^{ik \cdot (x-y)} \right]_{y=x} \\ &= -2 \int \frac{d^4 k}{(2\pi)^4} \text{Tr} \left\{ \gamma_5 t f(-[i\not{k} + \not{D}_x]^2/M^2) \right\}. \end{aligned}$$

(A derivative with respect to x gives zero when acting on the extreme right in the second expression, but not when acting on $A_{\alpha\mu}(x)$.) Rescaling the momentum k^μ by a factor M , this is

$$\mathcal{A}(x) = -2M^4 \int \frac{d^4 k}{(2\pi)^4} \text{Tr} \left\{ \gamma_5 t f(-[i\not{k} + \not{D}_x/M]^2) \right\}. \quad (22.2.17)$$

The argument of the cut-off function may be written

$$-\left[i\not{k} + \frac{\not{D}_x}{M}\right]^2 = k^2 - \frac{ik \cdot D_x}{M} - \left(\frac{\not{D}_x}{M}\right)^2. \quad (22.2.18)$$

Eq. (22.2.17) receives contributions in the limit $M \rightarrow \infty$ only from terms in the expansion of $f(-[i\not{k} + \not{D}_x/M]^2)$ that have no more than four factors of $1/M$, and also only from terms that contain at least four Dirac gamma matrices, because otherwise the trace over Dirac indices vanishes. This leaves us with only the terms of second order in $\not{D}_x^2$:

$$\mathcal{A}(x) = - \int \frac{d^4 k}{(2\pi)^4} f''(k^2) \text{Tr} \{ \gamma_5 t \not{D}_x^4 \}, \quad (22.2.19)$$

which are now independent of the regulator mass M .

To evaluate the integral over k , we perform the same sort of rotation of the k^0 contour of integration as in evaluating Feynman diagrams, so that in effect k^0 is replaced with ik^4 , with k^4 running from $-\infty$ to $+\infty$. (This is a step that can really only be justified by working with Euclidean path integrals from the beginning.) The integral is then in effect

$$\int d^4 k f''(k^2) = i \int_0^\infty 2\pi^2 \kappa^3 d\kappa f''(\kappa^2). \quad (22.2.20)$$

¹For instance, we might take $f(s) = e^{-s^2}$, as done originally by Fujikawa, or $f(s) = 1/(1+s)$.

By repeated integration by parts using Eq. (22.2.16) and then Eq. (22.2.15), this is

$$\int d^4k f''(k^2) = i\pi^2 \int_0^\infty ds s f''(s) = -i\pi^2 \int_0^\infty ds f'(s) = i\pi^2. \quad (22.2.21)$$

To calculate the trace, we write

$$\begin{aligned} \not{D}_x^2 &= \frac{1}{4} \{ (D_x)^\mu, (D_x)^\nu \} \{ \gamma_\mu, \gamma_\nu \} + \frac{1}{4} [(D_x)^\mu, (D_x)^\nu] [\gamma_\mu, \gamma_\nu] \\ &= D_x^2 - \frac{1}{4} i t_\alpha F_\alpha^{\mu\nu} [\gamma_\mu, \gamma_\nu]. \end{aligned} \quad (22.2.22)$$

The only term in $\not{D}_x^4$ that contributes to the trace over Dirac indices is the one involving a product of four Dirac matrices, which gives

$$\text{tr}_D \{ \gamma_5 [\gamma_\mu, \gamma_\nu] [\gamma_\rho, \gamma_\sigma] \} = 16i \epsilon_{\mu\nu\rho\sigma}, \quad (22.2.23)$$

where ‘ tr_D ’ denotes a trace over Dirac indices only, and as usual $\epsilon_{\mu\nu\rho\sigma}$ is the totally antisymmetric tensor with $\epsilon^{0123} = +1$. Using Eqs. (22.2.21)–(22.2.23) in Eq. (22.2.19) then gives the anomaly function as

$$\mathcal{A}(x) = -\frac{1}{16\pi^2} \epsilon_{\mu\nu\rho\sigma} F_\alpha^{\mu\nu}(x) F_\beta^{\rho\sigma}(x) \text{tr} \{ t_\alpha t_\beta t \}, \quad (22.2.24)$$

with ‘ tr ’ here denoting a trace only over indices labelling the various fermion species. In the special case in which t is the unit matrix, the quantity (22.2.24) is known as the *Chern–Pontryagin density*.

This result can be expressed in terms of the current associated with the anomalous symmetry. If we assume for simplicity that the action itself is invariant under the symmetry transformation $\psi(x) \rightarrow \psi(x) + it\gamma_5\alpha\psi(x)$ with constant infinitesimal parameter α , then as discussed in Section 7.3, the change in the action when we make such a transformation with a spacetime-dependent parameter $\alpha(x)$ may be written as $\delta I = \int d^4x J_5^\mu(x) \partial_\mu \alpha(x)$, where $J_5^\mu(x)$ is the current that becomes conserved when the field operators satisfy the dynamical equations that make the action stationary with respect to arbitrary variations in the fields. When we make the change of variables $\delta\psi(x) = it\gamma_5\alpha(x)\psi(x)$, the change in the path integral over the fermionic fields is

$$\delta \int [d\psi][d\bar{\psi}] e^{iI} = i \int d^4x \int [d\psi][d\bar{\psi}] [\mathcal{A}(x)\alpha(x) + J_5^\mu(x) \partial_\mu \alpha(x)] e^{iI}. \quad (22.2.25)$$

But this is a mere change of variables, and so for arbitrary $\alpha(x)$ it cannot affect the path integral. Therefore for arbitrary gauge fields

$$\langle \partial_\mu J_5^\mu \rangle_A = \mathcal{A} = -\frac{1}{16\pi^2} \epsilon_{\mu\nu\rho\sigma} F_\alpha^{\mu\nu} F_\beta^{\rho\sigma} \text{tr} \{ t_\alpha t_\beta t \}, \quad (22.2.26)$$

where for any operator $\mathcal{O}$, $\langle \mathcal{O} \rangle_A$ is the quantum average of $\mathcal{O}$ in a fixed background field $A^\mu(x)$:

$$\langle \mathcal{O} \rangle_A \equiv \frac{\int [d\psi][d\bar{\psi}] e^{iI} \mathcal{O}}{\int [d\psi][d\bar{\psi}] e^{iI}}. \quad (22.2.27)$$

Incidentally, it is possible to rewrite Eq. (22.2.26) as a conservation condition. Consider the special case where $\text{tr}\{t_\alpha t_\beta t\}$ is proportional to $\delta_{\alpha\beta}$:

$$\text{tr}\{t_\alpha t_\beta t\} = N\delta_{\alpha\beta}. \quad (22.2.28)$$

Define a current known as the *Chern-Simons class*:

$$\begin{aligned} G^\mu &\equiv 2\epsilon^{\mu\nu\lambda\rho} \left[A_{\gamma\nu} \partial_\lambda A_{\gamma\rho} + \frac{1}{3} C_{\alpha\beta\gamma} A_{\alpha\nu} A_{\beta\lambda} A_{\gamma\rho} \right] \\ &= \epsilon^{\mu\nu\lambda\rho} \left[A_{\gamma\nu} F_{\gamma\lambda\rho} - \frac{1}{3} C_{\alpha\beta\gamma} A_{\alpha\nu} A_{\beta\lambda} A_{\gamma\rho} \right], \end{aligned} \quad (22.2.29)$$

which satisfies the identity

$$\partial_\mu G^\mu = \frac{1}{2} \epsilon^{\mu\nu\lambda\rho} F_{\gamma\mu\nu} F_{\gamma\lambda\rho}. \quad (22.2.30)$$

Eq. (22.2.30) allows us to rewrite Eq. (22.2.26) as the condition

$$\partial_\mu K^\mu = 0, \quad (22.2.31)$$

where

$$K^\mu \equiv \langle J_5^\mu \rangle_A + \frac{N}{8\pi^2} G^\mu. \quad (22.2.32)$$

However, we cannot use the conservation of the current K^μ to argue, as we did in the previous section, that the process $\pi^0 \rightarrow 2\gamma$ is suppressed, because in making that argument we assumed not only the chiral symmetry associated with axial-vector current conservation but also electromagnetic gauge invariance, and as we see from Eq. (22.2.29), the current K^μ though conserved is not gauge-invariant.

Our derivation of the formula (22.2.24) for the anomaly shows that if we had evaluated the anomaly function using a differential operator $f(-\not{\partial}_x^2/M^2)$ in place of $f(-\not{D}_x^2/M^2)$ in Eq. (22.2.13), we would have obtained a vanishing anomaly function. In fact, with this regulator procedure the axial-vector current is K^μ , not J_5^μ . The trouble with this procedure, as mentioned above, is that the regulator operator is no longer gauge-invariant, resulting in the presence of a non-gauge-invariant term in K^μ . There is no regulator procedure for fermion propagators as well as determinants that is both gauge- and chiral-invariant.

We may now return to the problem that gave anomalies their start, and use our results to calculate the actual rate of the process $\pi^0 \rightarrow 2\gamma$. The symmetry of interest here is generated by charge-neutral chiral transformations of the light quark fields:

$$\delta u = i\alpha\gamma_5 u, \quad \delta d = -i\alpha\gamma_5 d. \quad (22.2.33)$$

This symmetry is anomaly-free in pure quantum chromodynamics, because the u and d belong to the same representation of the color gauge group, so that their contributions to the gluon-gluon terms in the anomaly of the symmetry (22.2.33) cancel. On the other hand, in the presence of an electromagnetic field $A^\mu(x)$, this symmetry has an anomaly

$$\mathcal{A}(x) = -\frac{1}{16\pi^2} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu}(x) F^{\rho\sigma}(x) \text{tr}\{q^2 \tau_3\},$$

where q is the quark charge matrix and τ_3 is the diagonal 2×2 matrix with elements $+1$ for u and -1 for d . If we assume as usual that there are N_c u quarks of charge $2e/3$ and an equal number of d quarks of charge $-e/3$, the trace is

$$\text{tr}\{q^2\tau_3\} = N_c \left(\frac{2e}{3}\right)^2 (+1) + N_c \left(\frac{-e}{3}\right)^2 (-1) = \frac{N_c e^2}{3},$$

so the anomaly function here is

$$\mathcal{A}(x) = -\frac{N_c e^2}{48\pi^2} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu}(x) F^{\rho\sigma}(x). \quad (22.2.34)$$

We must then include in the effective Lagrangian terms that under the chiral transformation (22.2.33) give it the transformation (22.2.12), that is,

$$\delta\mathcal{L}_{\text{eff}}(x) = \alpha\mathcal{A}(x) = -\frac{N_c e^2}{48\pi^2} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu}(x) F^{\rho\sigma}(x) \alpha. \quad (22.2.35)$$

Under the transformation (22.2.33) the pion field transforms as

$$\delta\pi^0 = \alpha F_\pi, \quad (22.2.36)$$

where $F_\pi = 184$ MeV is the pion decay amplitude introduced in Chapter 19. (The normalization convention for this constant is fixed by our definition of the symmetry generator as $\gamma_5\tau_3 = 2\gamma_5 t_3$.) It follows that we must include in the effective Lagrangian a term

$$\frac{\pi^0(x)\mathcal{A}(x)}{F_\pi} = -\frac{N_c e^2}{48\pi^2 F_\pi} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu}(x) F^{\rho\sigma}(x) \pi^0(x). \quad (22.2.37)$$

Comparing this with the general formula (22.1.1) for the effective Lagrangian for the decay $\pi^0 \rightarrow 2\gamma$, we see that the constant g in Eq. (22.2.1) must have the value

$$g = \frac{N_c e^2}{48\pi^2 F_\pi}. \quad (22.2.38)$$

(This shows that our previous crude order-of-magnitude estimate (22.1.3) was too large by just a factor $6/N_c$.) The rate (22.1.2) for pion decay is thus predicted to be

$$\Gamma(\pi^0 \rightarrow 2\gamma) = \frac{N_c^2 \alpha^2 m_\pi^3}{144\pi^3 F_\pi^2} = \left(\frac{N_c}{3}\right)^2 \cdot 1.11 \cdot 10^{16} \text{ s}^{-1}. \quad (22.2.39)$$

The observed rate is $\Gamma(\pi^0 \rightarrow 2\gamma) = (1.19 \pm 0.08) \cdot 10^{16} \text{ s}^{-1}$, in good agreement with the theoretical result (22.2.39) if and only if $N_c = 3$. The success of this calculation was one of the first pieces of firm evidence that there are three colors of quarks.

Remarkably, as we have seen in the previous section, shortly after the discovery of the π^0 Steinberger calculated g from a single proton-loop diagram, and obtained the result $g = e^2 G / 32\pi^2 m_N$, where G is the pseudoscalar pion–nucleon coupling constant. This result would precisely agree with Eq. (22.2.38) for $N_c = 3$ if we used the Goldberger–Treiman relation with $g_A = 1$ to set $G = 2m_N / F_\pi$. The correct result is larger than the Steinberger result by a factor $g_A^2 = 1.56$. The reason that Steinberger got nearly the right result is because the answer is determined by the triangle anomaly, which is proportional to $\text{tr}\{q^2\tau_3\}$. For one proton, this trace has the value e^2 , which is the same as the quark value found above for three colors of quarks.

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As mentioned earlier, a more rigorous derivation of the anomaly can be given by using path integrals in Euclidean spacetime. (The use of Euclidean path integrals is briefly discussed in [Appendix A of Chapter 23](#).) We introduce a Euclidean fourth coordinate $x_4 = ix^0 = -ix_0$, and correspondingly $\partial_4 = -i\partial_0$, $\gamma_4 = i\gamma^0$, and $A_{4\alpha} = iA_\alpha^0$. The spacetime volume is then expressed as $d^4x = -i(d^4x)_E$, where $(d^4x)_E$ is the Euclidean volume element $(d^4x)_E = dx_1 dx_2 dx_3 dx_4$. In Euclidean spacetime the fields $\psi(x)$ and $\bar{\psi}(x)$ must be regarded as entirely independent, with local chiral transformations defined by $\delta\psi(x) = i\alpha(x)t\gamma_5\psi(x)$, $\delta\bar{\psi}(x) = -i\alpha(x)\bar{\psi}(x)t\gamma_5$. The transformation of the measure is then again given by Eq. (22.2.10), with the anomaly function $\mathcal{A}(x)$ given by Eq. (22.2.11). Introducing a regulator function as before, this again gives the formula (22.12.13) for $\mathcal{A}(x)$. One great advantage of the Euclidean approach is that with x_4 and $A_{4\alpha}$ real, the Dirac operator $i\mathcal{D}$ in Eq. (22.2.13) is Hermitian:

$$i\mathcal{D} = [i\partial_i + t_\alpha A_{i\alpha}]\gamma_i, \quad (22.2.40)$$

with i, j , etc. here summed over the values 1, 2, 3, 4. It therefore has orthonormal spinor eigenfunctions $\varphi_\kappa(x)$:

$$i\mathcal{D}\varphi_\kappa = \lambda_\kappa\varphi_\kappa, \quad (22.2.41)$$

$$\int (d^4x)_E \varphi_\kappa(x)^\dagger \varphi_{\kappa'}(x) = \delta_{\kappa\kappa'}, \quad (22.2.42)$$

with real eigenvalues λ_κ . We also are assuming, as throughout this section, that t commutes with $i\mathcal{D}$, so that we can choose the φ_κ so that $t\varphi_\kappa = t_\kappa\varphi_\kappa$. These eigenfunctions satisfy a completeness relation

$$\sum_\kappa \varphi_\kappa(x) \varphi_\kappa^\dagger(y) = \delta^4(x - y) \mathbf{1}, \quad (22.2.43)$$

where ‘ $\mathbf{1}$ ’ is the $4 \cdot 4$ unit matrix. Therefore the anomaly function may now be written as the limit of a manifestly convergent sum:

$$\begin{aligned} \mathcal{A}(x) &= -2 \lim_{M \rightarrow \infty} \text{Tr} \left\{ \gamma_5 t f(-\mathcal{D}^2/M^2) \sum_\kappa \varphi_\kappa(x) \varphi_\kappa^\dagger(x) \right\} \\ &= -2 \lim_{M \rightarrow \infty} \sum_\kappa t_\kappa f(\lambda_\kappa^2/M^2) (\varphi_\kappa^\dagger(x) \gamma_5 \varphi_\kappa(x)). \end{aligned} \quad (22.2.44)$$

In just the same way that we derived formula (22.2.24) for the anomaly function, we can show here that

$$\mathcal{A}(x) = \frac{1}{16\pi^2} \epsilon_{ijkl}^E F_{ij\alpha} F_{kl\beta} \text{tr}\{t_\alpha t_\beta t\}, \quad (22.2.45)$$

where ϵ_{ijkl}^E is the totally antisymmetric tensor with $\epsilon_{1234}^E = +1$. (The difference of sign in Eqs. (22.2.24) and (22.2.45) arises because Eq. (22.2.45) is missing two factors of i as compared with Eq. (22.2.24): one factor i from Eq. (22.2.23), because

$$\text{tr}_D\{\gamma_5[\gamma_i, \gamma_j][\gamma_k, \gamma_l]\} = 16\epsilon_{ijkl}^E,$$

together with another factor i from the replacement of d^4k with $(d^4k)_E$ in Eq. (22.2.20).)

Now, given any eigenfunction $\varphi_\kappa(x)$ of $i\mathcal{D}$ and t with eigenvalue $\lambda_\kappa \neq 0$, there is another normalized eigenfunction $\varphi_{\kappa^-}(x)$ with eigenvalues $\lambda_{\kappa^-} = -\lambda_\kappa$ and t_{κ^-} , given by $\varphi_{\kappa^-}(x) = \gamma_5 \varphi_\kappa(x)$. (Recall that in the notation used throughout this book, γ_5 is the Hermitian matrix $\gamma_5 = -i\gamma_1\gamma_2\gamma_3\gamma_0 = \gamma_1\gamma_2\gamma_3\gamma_4$.) Furthermore $(\varphi_\kappa^\dagger(x)\varphi_\kappa(x)) = (\varphi_{\kappa^-}^\dagger(x)\varphi_{\kappa^-}(x))$, so the terms κ and κ^- cancel in the sum in Eq. (22.2.44). This leaves just a sum over the eigenfunctions with $\lambda_\kappa = 0$. These are not generally paired; rather, since γ_5 anticommutes with $i\mathcal{D}$, they can be chosen as simultaneous orthonormal eigenfunctions φ_u, φ_v of $i\mathcal{D}$ with eigenvalue zero and of γ_5 with eigenvalues $+1$ and -1 , respectively.

$$\begin{aligned} i\mathcal{D}\varphi_u &= 0, & \gamma_5\varphi_u &= \varphi_u, \\ i\mathcal{D}\varphi_v &= 0, & \gamma_5\varphi_v &= -\varphi_v. \end{aligned} \quad (22.2.46)$$

Using the fact that $f(0) = 1$, Eq. (2.2.44) then becomes

$$\mathcal{A}(x) = -2 \left[\sum_u t_u (\varphi_u^\dagger(x)\varphi_u(x)) - \sum_v t_v (\varphi_v^\dagger(x)\varphi_v(x)) \right]. \quad (22.2.47)$$

Now, since the φ_u and φ_v are normalized as in Eq. (22.2.42), the integral of Eq. (22.2.47) gives

$$\int (d^4x)_E \mathcal{A}(x) = -2 \left[\sum_u t_u - \sum_v t_v \right], \quad (22.2.48)$$

with the sums over u and v running over left- and right-handed zero modes of the operator $i\mathcal{D}$, respectively. In particular, for the case where t is the unit matrix, by using Eq. (22.2.45) this may be expressed as a relation between a functional of the gauge field and the numbers of the zero modes of the Dirac operator with definite chiralities:

$$-\frac{1}{32\pi^2} \int (d^4x)_E \epsilon_{ijkl}^E F_{\alpha ij} F_{\beta kl} \text{tr}[t_\alpha t_\beta] = n_+ - n_-, \quad (22.2.49)$$

where here $n_\pm$ are the numbers of zero modes of $\mathcal{D}$ that have eigenvalues ± 1 for γ_5 . This is the celebrated *Atiyah–Singer index theorem*. [6a] Among other things, it shows that under variations in the gauge field the integral on the left-hand side of Eq. (22.2.49) cannot change smoothly, but only by integers, so this integral can only depend on the topology of the gauge field. Its dependence on this topology will be described in Section 23.5.